

Summer School Project Notes

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1 The Wave Equation

Let $t \geq 0$ denote the time coordinate. Let $L > 0$ denote a length unit. Let $x \in [-L, L]$ denote the spatial coordinate. Let $c > 0$ denote a wave speed. Let $u^0(x), v^0(x) \in \mathbb{R}$ be periodic functions of x , and let $f(x, t) \in \mathbb{R}$ be a function of x, t that is periodic in x . We seek a function $u(x, t) \in \mathbb{R}$ of x and t that is periodic in x that solves the initial value problem for the wave equation:

$$\partial_{tt}u(x, t) - c^2\partial_{xx}u(x, t) = f(x, t), \quad (1)$$

$$u(x, 0) = u^0(x), \quad (2)$$

$$\partial_t u(x, 0) = v^0(x). \quad (3)$$

We assume that the data u^0, v^0, f are such that there exists a unique solution u to this problem. We will develop a finite difference technique to approximate u .

1.1 Discretization

We let $k \geq 0$ denote the refinement level. We create a spatial mesh of $[-L, L]$ by subdividing it into $M := 2^k$ subintervals of mesh size $h := 2L/M$. We let $x_i := -L + ih$, $0 \leq i \leq M$ denote the grid points of our mesh. We let $\tau > 0$ denote the time step size, chosen so that the CFL condition $c\tau < h$ holds. We let $t_n := n\tau$, $n \geq 0$ denote the n th discrete time point. We let $U_i^n \in \mathbb{R}$ denote the finite difference approximation to $u(t_n, x_i)$. We let $f_i^n := f(t_n, x_i)$, $u_i^0 := u^0(x_i)$, $v_i^0 := v^0(x_i)$.

We use second-order centered difference stencils in time and space. This gives us

$$\frac{U_i^{n+1} - 2U_i^n + U_i^{n-1}}{\tau^2} + c^2 \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{h^2} = f_i^n \quad (4)$$

for $1 \leq i \leq M-1$ and $n \geq 1$. For $n \geq 1$, we incorporate the periodic boundary conditions by imposing

$$\frac{U_0^{n+1} - 2U_0^n + U_0^{n-1}}{\tau^2} + c^2 \frac{U_1^n - 2U_0^n + U_{M-1}^n}{h^2} = f_0^n, \quad (5)$$

$$U_M^n = U_0^n. \quad (6)$$

For initial conditions, we set

$$U_i^0 = u_i^0, \quad (7)$$

$$U_i^1 = U_i^0 + \tau v_i^0 + \frac{c^2 \tau^2}{2h^2} (U_{i+1}^0 - 2U_i^0 + U_{i-1}^0) + \frac{\tau^2}{2} f_i^0 \quad (8)$$

for $0 \leq i \leq M$. Then, for $n \geq 1$, $0 \leq i \leq M$, we update U_i^{n+1} via (4), (5), or (6).

1.2 Exercises

1. Implement the finite difference scheme above for the wave equation with periodic boundary conditions.
2. Test your code with the method of manufactured solutions. Let $L = 10$, $c = 1$, $u(t, x) = \exp(-(x - ct)^2)$. Compute the corresponding f , u^0 , and v^0 such that u solves the PDE. Run until final time $T := L/(2c) = N\tau$. Plot U_i^N versus $u(t_N, x_i)$ at the grid points x_i for the highest refinement level.

3. Let

$$e_k := \max_{0 \leq i \leq M} |U_i^N - u(t_N, x_i)| \quad (9)$$

denote the discrete error at the final time for mesh refinement level k , where we recall that $M = 2^k$. Make a table that displays the mesh size, the error, and the rate. Create a log-log plot of e_k versus the mesh size $h_k := 2L/M = L/2^{k-1}$ for $k = 10, 11, 12$.

4. Make an animation of your approximate solution U_i^n .
5. Now set $f(t, x) = 0$ and $v^0 = 0$. Experiment with different periodic initial conditions u^0 . Take snapshots of your simulations and create animations over the circle in 3d.

Since we are modeling periodic solutions, we can plot them over a circle of circumference $2\pi r = 2L$ in 3d in the following way. We let $s \in [-L, L]$ denote our spatial coordinate, so that our discrete grid points are denoted by s_i , and U_i^n denotes the approximation to $u(t_n, s_i)$. Then, s corresponds to an arc length coordinate $s = r\theta$ on the circle of radius $r = L/\pi$. By making the transformation

$$s \mapsto \left(r \cos\left(\frac{s}{r}\right), r \sin\left(\frac{s}{r}\right) \right), \quad (10)$$

we can map our 2d points

$$(s_i, U_i^n) \mapsto \left(r \cos\left(\frac{s_i}{r}\right), r \sin\left(\frac{s_i}{r}\right), U_i^n \right), \quad (11)$$

to obtain a plot of our discrete solution over the circle in 3d.

2 Dispersive Wave Equations

We modify the wave equation by adding an additional term that depends on u :

$$\partial_t^2 u - c^2 \partial_x^2 u + g(u) = f(t, x), \quad (12)$$

$$u(0, x) = u^0(x), \quad (13)$$

$$\partial_t u(0, x) = v^0(x). \quad (14)$$

Example 1 (Klein–Gordon). Choosing $g(u) = u$, $f(t, x) = 0$ gives us the Klein–Gordon equation. This is a linear dispersive PDE, where the wave speed of the solutions depends on its frequency. This can be seen by the following computation. Assume that $u(t, x) = A \exp(i(kx - \omega t))$ is a plane wave solution to the Klein–Gordon equation with wavenumber k frequency ω , and amplitude A . Inserting this into the PDE gives us the relation

$$-\omega^2 + c^2 k^2 + 1 = 0, \quad (15)$$

which is called the dispersion relation for the Klein–Gordon equation. Dividing both sides by k^2 and rearranging gives us

$$v^2 = c^2 + \lambda^2, \quad (16)$$

where $v := \omega/k$ is the wave velocity and $\lambda := 1/k$ is the wavelength. This dispersion relation tells us that plane wave solutions of the Klein–Gordon equation with different wavelengths travel at different velocities, with shorter wavelengths traveling slower than longer wavelengths. In contrast, the dispersion relation for the wave equation with $g(u) = 0$, $f(t, x) = 0$ is

$$v^2 = c^2, \quad (17)$$

which means that all plane wave solutions to the wave equation travel at the same speed c .

For the general case, we decompose a wave solution u of the Klein–Gordon equation into its fundamental frequencies

$$u(t, x) = \sum_{k=-\infty}^{\infty} A_k e^{ik(x-v_k t)} + B_k e^{ik(x+v_k t)}, \quad (18)$$

with $v_k^2 = c^2 + \lambda_k^2$, $\lambda_k = 1/|k|$. As time evolves, the plane waves with wavelength λ_k will travel faster than the plane waves with wavelength λ_j for $j \leq k$, causing the initial profile of the solution $u(0, x)$ to spread out (i.e. disperse) over time. $\square$

2.1 Discretization

We use the scheme we developed for the wave equation with a small modification for the nonlinear term $g(u)$:

$$\frac{U_i^{n+1} - 2U_i^n + U_i^{n-1}}{\tau^2} + c^2 \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{h^2} + g(U_i^n) = f_i^n \quad (19)$$

We terminate all simulations when the wave fronts reach the boundary of the domain, so the choice of boundary condition does not matter much for our simulations. They only need to give a stable and well-posed linear system at the discrete level. The periodic boundary conditions from the previous section accomplish this. Another option is to use an absorbing boundary condition. At the lowest order approximation, this amounts to simply setting

$$U_0^{n+1} = U_1^n, \quad (20)$$

$$U_M^{n+1} = U_{M-1}^n \quad (21)$$

for $n \geq 1$.

2.2 Exercises

1. Extend the finite difference scheme for the wave equation to allow for $f(t, x, u)$ and the ability to choose either periodic or absorbing boundary conditions.
2. Test your code with the method of manufactured solutions. Let $L = 100$, $c = 1$, $g(u) = \sin(u)$,

$$u(t, x) = 4 \arctan \left(\frac{\cos(\omega t) \sqrt{1 - \omega^2}}{\omega \cosh(x \sqrt{1 - \omega^2})} \right). \quad (22)$$

with $\omega = 0.5$. Compute the corresponding f , u^0 , and v^0 such that u solves the PDE. Run until final time $T := L/c$. Let

$$e_k := \max_{0 \leq i \leq M} |U_i^N - u(t_N, x_i)| \quad (23)$$

denote the discrete error at the final time for mesh refinement level k , where we recall that $M = 2^k$. Create a log-log plot of e_k versus the mesh size $h_k := 2L/M = L/2^{k-1}$ for $k = 10, 11, 12$. Plot U_i^N versus $u(t_N, x_i)$ at the grid points x_i for the highest refinement level.

3. Repeat the previous exercise, but now for

$$u(t, x) = 4 \arctan \left(\frac{\cos(\omega \gamma t - \omega x \sqrt{\gamma^2 - 1}) \sqrt{1 - \omega^2}}{\omega \cosh((\gamma x - t \sqrt{\gamma^2 - 1}) \sqrt{1 - \omega^2})} \right). \quad (24)$$

with $\gamma = 1/\sqrt{1 - v^2}$ and $v = 1/\sqrt{2}$.

4. Fix $u^0(x) = \exp(-x^2)$, $v^0(x) = 0$, $c = 1$. Solve the PDE on $[-L, L]$ with absorbing boundary conditions for different choices of f :
 - (a) $g(u) = 0$, $f(t, x) = 0$ (Wave)
 - (b) $g(u) = u$, $f(t, x) = 0$ (Klein-Gordon)

- (c) $g(u) = \sin(u)$, $f(t, x) = 0$ (Sine–Gordon)
- (d) Whatever other conditions you want to test, either by finding it in the literature or thinking of it yourself.

Make animations for each of the cases up to the same final time T , chosen so that the simulation terminates when the wave front hits the boundary. Experiment with different choices of L until you find one that gives interesting results. Use the same choice of L for each of the simulations.

3 Solitons

Certain nonlinear dispersive PDEs exhibit special solutions called solitons. They are characterized as localized waves that do not lose energy and which maintain their shape after collision with other soliton waves. We will model some soliton solutions to the sine–Gordon equation where $g(u) = 0$ and $f(t, x) = 0$:

1. Stationary breather:

$$u(t, x) = 4 \arctan \left(\frac{\cos(\omega t) \sqrt{1 - \omega^2}}{\omega \cosh(x \sqrt{1 - \omega^2})} \right). \quad (25)$$

2. Traveling breather:

$$u(t, x) = 4 \arctan \left(\frac{\cos \left(\omega \gamma t - \omega x \sqrt{\gamma^2 - 1} \right) \sqrt{1 - \omega^2}}{\omega \cosh \left(\left(\gamma x - t \sqrt{\gamma^2 - 1} \right) \sqrt{1 - \omega^2} \right)} \right). \quad (26)$$

3. Kink:

$$u(t, x) = 4 \arctan \left(\exp \left(\frac{x - vt}{\sqrt{1 - v^2}} + \delta \right) \right) \quad (27)$$